Project Assessment: Hypertension Risk Prediction

Center for Data Analytics and Modeling (CDAM)

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Instructions to Candidates:

1. Complete **ALL** tasks using Python. You will present your work on 8th June 2026 using Power Presentation (8-12 slides).
2. All code must be executable, well-commented, and reproducible (set `random_state=42` or `2026` where applicable).
3. Include brief markdown explanations or code comments interpreting outputs, justifying methodological choices, and addressing clinical relevance.
4. The dataset provided is `htn_dat.csv` (Hypertension Screening Data).

Project Context and Dataset Description

Project Objective: Develop and compare multiple predictive models to identify patients at risk of elevated systolic blood pressure (SBP $\geq$ 120 mmHg).

Dataset: `htn_dat.csv` contains patient records with the following variables:

Variable	Type	Description
`ID`	Integer	Unique patient identifier
`DBP`	Numeric	Diastolic Blood Pressure (mmHg)
`SBP`	Numeric	Systolic Blood Pressure (mmHg)
`BMI`	Numeric	Body Mass Index (kg/m^2)
`age`	Numeric	Age in years
`married`	Binary	Marital status (0 = No, 1 = Yes)
`male.gender`	Binary	Gender (0 = Female, 1 = Male)
`hgb_centered`	Numeric	Hemoglobin level (centered)
`adv_HIV`	Categorical	Advanced HIV status
`survtime`	Numeric	Survival/follow-up time
`event`	Binary	Event indicator (0/1)
`arv_naive`	Binary	ARV treatment naivety (0/1)

Variable	Type	Description
`urban.clinic`	Binary	Clinic location (0 = Rural, 1 = Urban)
`log_creat_centered`	Numeric	Log creatinine (centered)
`IPW_weight`	Numeric	Inverse probability weight
`SBP_ge120`	**Binary (Target)**	**1 = SBP $\geq$ 120 mmHg, 0 = SBP $<$ 120 mmHg**

C1. Data Understanding and Preparation (10 Marks)

(a) Write Python code to load `htn_dat.csv`, perform initial inspection (`info()`, `describe()`, `shape`), and generate **FOUR** distinct exploratory analyses or visualizations. Examples include: target class distribution plot, missing value heatmap, numerical feature distributions (histograms/boxplots), or a correlation matrix heatmap. Include brief comments interpreting key patterns. (5 marks)

(b) The target variable `SBP_ge120` is binary. Implement the following in Python:

- (i) Calculate and print the class distribution and imbalance ratio. Explain in a comment why imbalance matters for modeling and metric selection. (2 marks)
- (ii) Handle missing values in predictor variables using an appropriate strategy (e.g., median/mode imputation, or model-based imputation). Justify your choice with inline comments referencing medical data characteristics. (2 marks)
- (iii) Encode categorical/binary variables (e.g., `urban.clinic`, `adv_HIV`) so they are compatible with scikit-learn models. Show the transformation code. (1 mark)

C2. Model Development Strategy (11 Marks)

(a) Split the preprocessed data into training and testing sets. Provide code and justify your choices in comments:

- (i) Split ratio (e.g., 70:30 or 80:20).
- (ii) Use of stratified sampling.
- (iii) Setting a fixed random state for reproducibility.

(4 marks)

(b) Build and train **SEVEN** classification models using `scikit-learn`: Binary Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Naive Bayes, Decision Tree, Random Forest, and Extreme Gradient Boosting (XGBoost).

- (i) Apply `StandardScaler` *only* where algorithmically required (e.g., LR, KNN, SVM) using `Pipeline` or consistent preprocessing.
- (ii) Explicitly set at least **two key hyperparameters** per model.

(iii) Add inline comments for each model explaining: (1) why scaling is/isn't required, (2) one practical advantage, and (3) one limitation for clinical hypertension prediction.

(7 marks)

C3. Model Evaluation and Comparison (11 Marks)

(a) Generate predictions and probability scores on the test set for all seven models. Calculate and display a comparison DataFrame of at least **FOUR** evaluation metrics (e.g., Accuracy, Precision, Recall/Sensitivity, F1-Score, ROC-AUC, Specificity). In comments, explain why each selected metric is clinically relevant for hypertension screening. (7 marks)

(b) Implement `StratifiedKFold` cross-validation to obtain stable performance estimates:

(i) Compute mean and standard deviation of ROC-AUC across folds for each model.

(ii) Plot all **seven** models' ROC curves on a single figure with a legend displaying AUC values.

(iii) Add a comment explaining how cross-validation improves reliability over a single train-test split in healthcare applications.

(4 marks)

C4. Interpretation and Recommendations (8 Marks)

(a) Extract feature importances from the trained Random Forest (or XGBoost) model. Plot the top 10 features. Identify the top 3 predictors and write a concise markdown cell or commented block interpreting their clinical relevance. Suggest **actionable screening or intervention strategies** healthcare providers could derive from each predictor. (4 marks)

(b) Compare Binary Logistic Regression, Random Forest, and XGBoost based on your results. Write a structured recommendation (in markdown or detailed comments) selecting the most suitable model for deployment in a resource-limited Kenyan clinic. Justify your choice considering:

(i) Predictive performance (AUC, generalization)

(ii) Clinical interpretability (odds ratios vs. black-box predictions)

(iii) Computational requirements and inference speed

(iv) Ease of implementation, monitoring, and maintenance

(4 marks)

END OF ASSESSMENT

Good Luck!